Whitefield POP Huawei Access Node WFD-ACC-09 Routing Protocol Flap Resolution SOP

Node: WFD-ACC-09 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_1013

Description

This Standard Operating Procedure (SOP) addresses the failure domain of Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) on Huawei Access Node WFD-ACC-09 located at the Whitefield POP, which is a critical juncture in the telecommunications network architecture. The Huawei Access Node serves as a gateway between the customer's network and the service provider's network, facilitating the exchange of routing information and packet forwarding between the two domains. The routing protocols employed in this scenario include BGP, OSPF, and ISIS, which are responsible for maintaining the routing table and ensuring the efficient exchange of traffic between the various network segments. In the event of a routing protocol flap, the node's ability to maintain optimal routing and packet forwarding is severely compromised, leading to potential service disruptions and decreased network performance. The root cause of this failure domain lies in the complex interplay between the routing protocols and the node's hardware and software components, which can be triggered by factors such as misconfiguration, hardware failures, or software glitches.

Pre-requisite Checks

- check: Verify the Whitefield POP's accessibility and ensure that the site can be physically accessed for any necessary maintenance or repairs., priority: High

- check: Confirm the status of the redundant links connected to the Huawei Access Node and ensure that they are operational., priority: Medium

- check: Check the node's power supply and ensure that it is stable and within normal operating parameters., priority: Low

Potential Root Causes

Cause	Probability
Misconfiguration of BGP, OSPF, or ISIS protocols leading to routing table inconsistencies or adjacency issues.	High
Hardware failures or malfunctions within the node's routing engine or line cards.	Medium
Software glitches or bugs within the node's operating system or routing protocol software.	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp process | include neighbor, Expected Output: Neighbor AS Port State Up/Down Last Inq MsgRcvd MsgSent OutQ ConnDroprc, Description: This command is used to display the BGP neighbor information, including the state of the neighbor connections and any potential issues that may be contributing to the routing protocol flap.
```

```
Command: display ospf routing-table | include interface, Expected Output: Interface State Type Address Age Next-Hop, Description: This command is used to display the OSPF routing table, including any potential issues with interface connectivity or adjacency loss that may be contributing to the routing protocol flap.

Command: display isis database | include lsp, Expected Output: LSPID Seq Checksum Len Time Age, Description: This command is used to display the ISIS database, including any potential issues with LSP flooding or adjacency loss that may be contributing to the routing protocol flap.
```

Resolution Steps

Step 1: Step: Clear BGP, OSPF, or ISIS neighbor connections, Command: clear bgp *, Description: This step involves clearing the BGP neighbor connections to reset the routing table and allow the protocols to re-converge. The * symbol is used to clear all BGP connections.

Step 2: Step: Verify OSPF or ISIS adjacency and adjust as necessary, Command: display ospf routing-table | exclude interface, Description: This step involves verifying the OSPF or ISIS adjacency and making any necessary adjustments to the interface configuration to ensure optimal routing and packet forwarding.

Step 3: Step: Verify ISIS LSP flooding and adjust as necessary, Command: display isis database | exclude lsp, Description: This step involves verifying the ISIS LSP flooding and making any necessary adjustments to the LSP configuration to ensure optimal routing and packet forwarding.

Post-Check Verification

```
Command: display bgp process | include neighbor, Expected Output: Neighbor AS Port State Up/Down Last Inq MsgRcvd MsgSent OutQ ConnDroprc, Description: This command is used to verify that the BGP neighbor connections have been successfully cleared and re-established.

Command: display ospf routing-table | include interface, Expected Output: Interface State Type Address Age Next-Hop, Description: This command is used to verify that the OSPF or ISIS adjacency has been successfully established and is functioning properly.

Command: display isis database | include lsp, Expected Output: LSPID Seq Checksum Len Time Age, Description: This command is used to verify that the ISIS LSP flooding is functioning properly and is not contributing to the routing protocol flap.
```

Roll-back Procedures

- Step: Restore original BGP, OSPF, or ISIS configuration, Command: undo clear bgp *, Description: This step involves restoring the original BGP, OSPF, or ISIS configuration to its previous state, effectively rolling back any changes made during the resolution process.

- Step: Restore original interface configuration, Command: undo interface, Description: This step involves restoring the original interface configuration to its previous state, effectively rolling back any changes made during the resolution process.

- Step: Restore original LSP configuration, Command: undo lsp, Description: This step involves restoring the original LSP configuration to its previous state, effectively rolling back any changes made during the resolution process.